

Curriculum Vitae Hojin KIM

Contact

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Education

Ph.D. in Mathematical Sciences

KAIST, 2025

Advisor: Bo-Hae Im

Thesis: On arithmetic properties of period polynomials of modular forms

*Alternative civilian service as AI Researcher
on leave from Ph.D.*

*TmaxData & TmaxAI, Seongnam, South Korea
Sep. 2018 – Aug. 2020*

M.S. in Mathematical Sciences

KAIST, 2016

Advisor: Suh Hyun Choi

Thesis: Weil conjectures for elliptic curves

B.S. in Mathematical Sciences

KAIST, 2014

Employment

Postdoctoral researcher (upcoming) Jan. 2027 –
VIASM, Vietnam

Postdoctoral researcher Apr. 2026 – Aug. 2026
Chonnam National University, South Korea
Host: Jinha Kim

Postdoctoral researcher Feb. 2025 – Feb. 2026
LMNO, Université de Caen Normandie, France
Mentor: Tuan Ngo Dac

Research Interests

Number theory: algebraic structures of the multiple zeta values in positive characteristic; zeros and unimodularity phenomena of period polynomials of modular forms

List of All Publications

Journal Publications

- *Zagier-Hoffman's conjectures in positive characteristic* (Bo-Hae Im, **Hojin Kim**, Khac Nhuan Le, Tuan Ngo Dac, Lan Huong Pham), **Forum Math. Pi.** (2024) doi:10.1017/fmp.2024.18

- *On the common zeros of quasi-modular forms for $\Gamma_0^+(N)$ of level $N = 1, 2, 3$* (Bo-Hae Im, **Hojin Kim**, Wonwoong Lee), **Open Math.** 22 (2024), no. 1, 20240065. doi:10.1515/math-2024-0065
- *Note on the Linear Independence of Alternating Multiple Zeta Values in Positive Characteristic* (Bo-Hae Im, **Hojin Kim**, Khac Nhuan Le, Tuan Ngo Dac, Lan Huong Pham), **Acta Math. Vietnam.** (2024), 1–37. doi:10.1007/s40306-024-00554-4
- *Riemann hypothesis for period polynomials attached to the derivatives of L -functions of cusp forms for $\Gamma_0(N)$* (Bo-Hae Im, **Hojin Kim**), **J. Math. Anal. Appl.** 509 (2022), no. 2, Paper No. 125971. doi:10.1016/j.jmaa.2021.125971

Preprints

- *The threshold for linear independence of multiple zeta values in positive characteristic* (Bo-Hae Im, **Hojin Kim**, Tuan Ngo Dac), submitted, 2026. arXiv:2604.25451
- *Zagier-Hoffman’s conjectures in positive characteristic II* (Bo-Hae Im, **Hojin Kim**, Khac Nhuan Le, Tuan Ngo Dac, Lan Huong Pham), submitted, 2024. arXiv:2402.11539
- *Hopf algebras and alternating multiple zeta values in positive characteristic* (Bo-Hae Im, **Hojin Kim**, Khac Nhuan Le, Tuan Ngo Dac, Lan Huong Pham), submitted, 2023. arXiv:2304.02337
- *Hopf algebras and multiple zeta values in positive characteristic* (Bo-Hae Im, **Hojin Kim**, Khac Nhuan Le, Tuan Ngo Dac, Lan Huong Pham), submitted, 2023. arXiv:2301.05906

Talks

Contributed and Invited Talks

23 June 2026

$\mathbb{F}_q$ -linear relations among the multiple zeta values in positive characteristic.
Special Session on Number Theory with Analytic Flavor, *International Conference for the 80th Anniversary of the Korean Mathematical Society*, 22–25 June. Seoul, Korea.

11 August 2025

A linear independence of the multiple zeta values in positive characteristics.
KAIST Number Theory Seminar. Daejeon, Korea.

15 July 2025

A linear independence of the multiple zeta values in positive characteristics.
The 5th Korea-France Conference in Mathematics, 15–18 July. Bordeaux, France.

30 June 2025

A linear independence of the multiple zeta values in positive characteristic.
33èmes Journées Arithmétiques, 30 June–4 July. Esch-sur-Alzette, Luxembourg.

20 June 2025

A linear independence of the multiple zeta values in positive characteristic.
Rencontres arithmétiques de Caen : aspects p -adiques et modulo p , 18–20 June. Caen, France.

22 May 2025

Algebraic structures of Multiple Zeta Values in positive characteristics.
KIAS Number Theory Seminar. Online.

8 November 2024

Hopf algebra structures of Multiple Zeta Values in positive characteristics.
Workshop, *School and Workshop “Selected topics in Arithmetic Algebraic Geometry”*, 28 Oct.–8 Nov. Hanoi, Vietnam.

19 October 2022

Riemann hypothesis for period polynomials attached to the derivatives of L -functions of cusp forms for $\Gamma_0(N)$.

Special session on Automorphic Forms and q -Series, *2022 Global KMS International Conference*, 18–21 Oct. Seoul, Korea.

Other talks

19 June 2026

On the Disproof of the Unit Distance Conjecture.

Expository talk given at the GIST seminar, at the invitation of Prof. Minki Kim, on the 2026 result in discrete geometry produced by an OpenAI LLM, in which algebraic number theory of number fields plays a central role. The talk was based on arXiv:2605.20695, a paper not authored by the speaker.

Teaching Experience

TA for the following courses (underlined semesters for the Teaching Assistant Excellence Award).

Calculus I (2015 Fall, 2021 Spring, 2022 Spring, 2023 Spring)

Calculus II (2014 Spring, 2016 Fall, 2017 Fall, 2020 Fall, 2021 Fall, 2022 Fall)

Differential Equations and Applications (2016 Spring, 2017 Spring)

Logic and Set Theory (2014 Fall, 2016 Fall)

Introduction to Linear Algebra (2021 Spring)

Linear Algebra (2017 Spring, 2017 Fall, 2020 Fall, 2021 Fall, 2022 Fall)

Introduction to Number Theory (2015 Spring, 2016 Spring, 2022 Spring, 2023 Spring)

Awards and Honors

Teaching Assistant Excellence Award (3 times)

Sep. 2023, Aug. 2022, July 2021

Department of Mathematical Sciences, KAIST

Bronze Medal, Math Competition for College Students

Nov. 2011

Korean Mathematical Society

National Excellence Scholarship (Natural Sciences and Engineering)

Feb. 2009 – Feb. 2013

National Government of the Republic of Korea

Languages

I speak Korean (native), English (fluent), French and Spanish (basic).

I can code in Python, Java, Mathematica, SageMath, and $\text{\LaTeX}$.

Updated on June 25, 2026.